Joining data: a real-world necessity

PANDAS JOINS FOR SPREADSHEET USERS

John Miller
Principal Data Scientist

Pandas for spreadsheet users

- Learn based on similarities to spreadsheets
- Understand the power and flexibility of pandas
- Use data from the National Football League (NFL)

Common situations

- Datasets split by time or other factor
- Datasets with related factors

Split data

- Influenced by reporting cycle
- Common splits
  - Time
  - Geography
  - Business unit

Split data example

games_2016

GameKey	Season	SeasonType	HomeTeam
1	2016	Pre	Indianapolis Colts
2	2016	Pre	Los Angeles Rams
3	2016	Pre	Baltimore Ravens
4	2016	Pre	Green Bay Packers

games_2017

GameKey	Season	SeasonType	HomeTeam
334	2017	Pre	Dallas Cowboys
335	2017	Pre	Arizona Cardinals
336	2017	Pre	Baltimore Ravens
337	2017	Pre	Buffalo Bills

Split data example

GameKey	Season	SeasonType	HomeTeam
1	2016	Pre	Indianapolis Colts
2	2016	Pre	Los Angeles Rams
3	2016	Pre	Baltimore Ravens
4	2016	Pre	Green Bay Packers

games_2016

GameKey	Season	SeasonType	HomeTeam
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335	2017	Pre	Arizona Cardinals
336	2017	Pre	Baltimore Ravens
337	2017	Pre	Buffalo Bills

games_2017

Split data example

games_2016

GameKey	Season	SeasonType	HomeTeam
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2	2016	Pre	Los Angeles Rams
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4	2016	Pre	Green Bay Packers

games_2017

GameKey	Season	SeasonType	HomeTeam
334	2017	Pre	Dallas Cowboys
335	2017	Pre	Arizona Cardinals
336	2017	Pre	Baltimore Ravens
337	2017	Pre	Buffalo Bills

Complementary data

- Results from collecting data for different purposes
- Department-specific data
- Storage in separate files or database tables

Complementary data example

	A	B	C
1	GameKey	Season	SeasonType
2	1	2016	Pre
3	2	2016	Pre
4	3	2016	Pre
5	4	2016	Pre

	A	B	C
1	GameKey	Turf	Stadium
2	1	Turf	Tom Benson Hall of F
3	2	Grass	Los Angeles Memori
4	3	Natural Grass	M&T Bank Stadium
5	4	DD GrassMaster	Lambeau Field

Complementary data example

	A	B	C
1	GameKey	Season	SeasonType
2	1	2016	Pre
3	2	2016	Pre
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	A	B	C
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Let's practice!

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Concatenation

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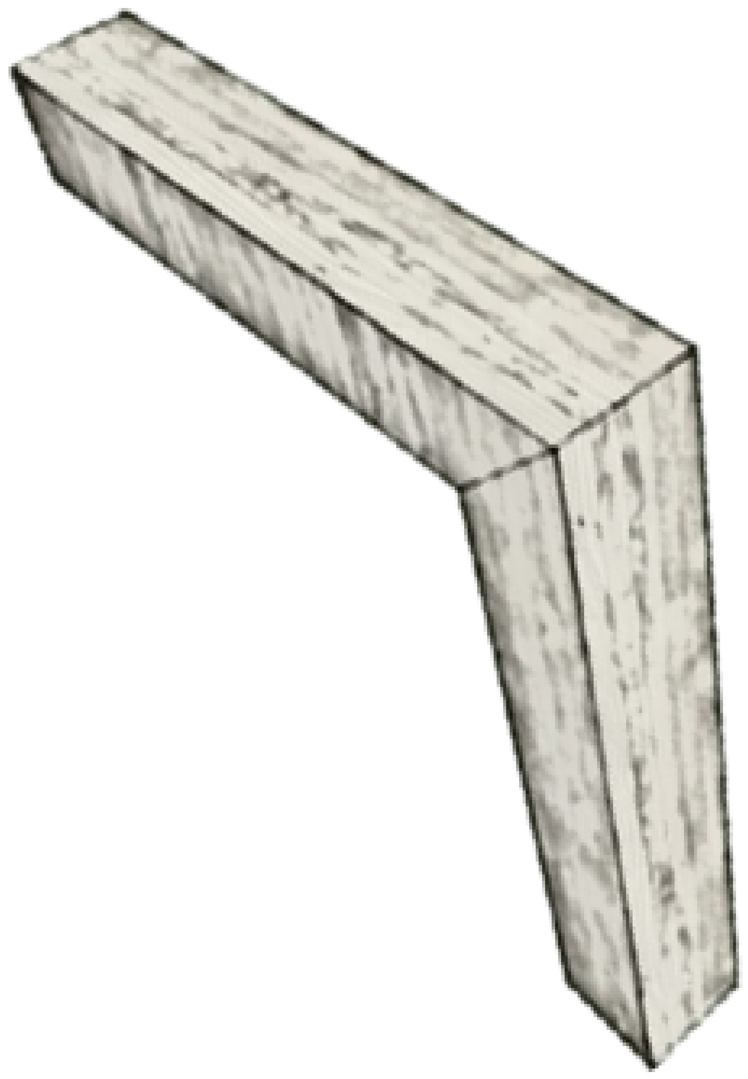
Concatenation basics

- Similar to spreadsheet CONCATENATE
- Mimics copy-paste of cells
- `pd.concat()` along rows or columns

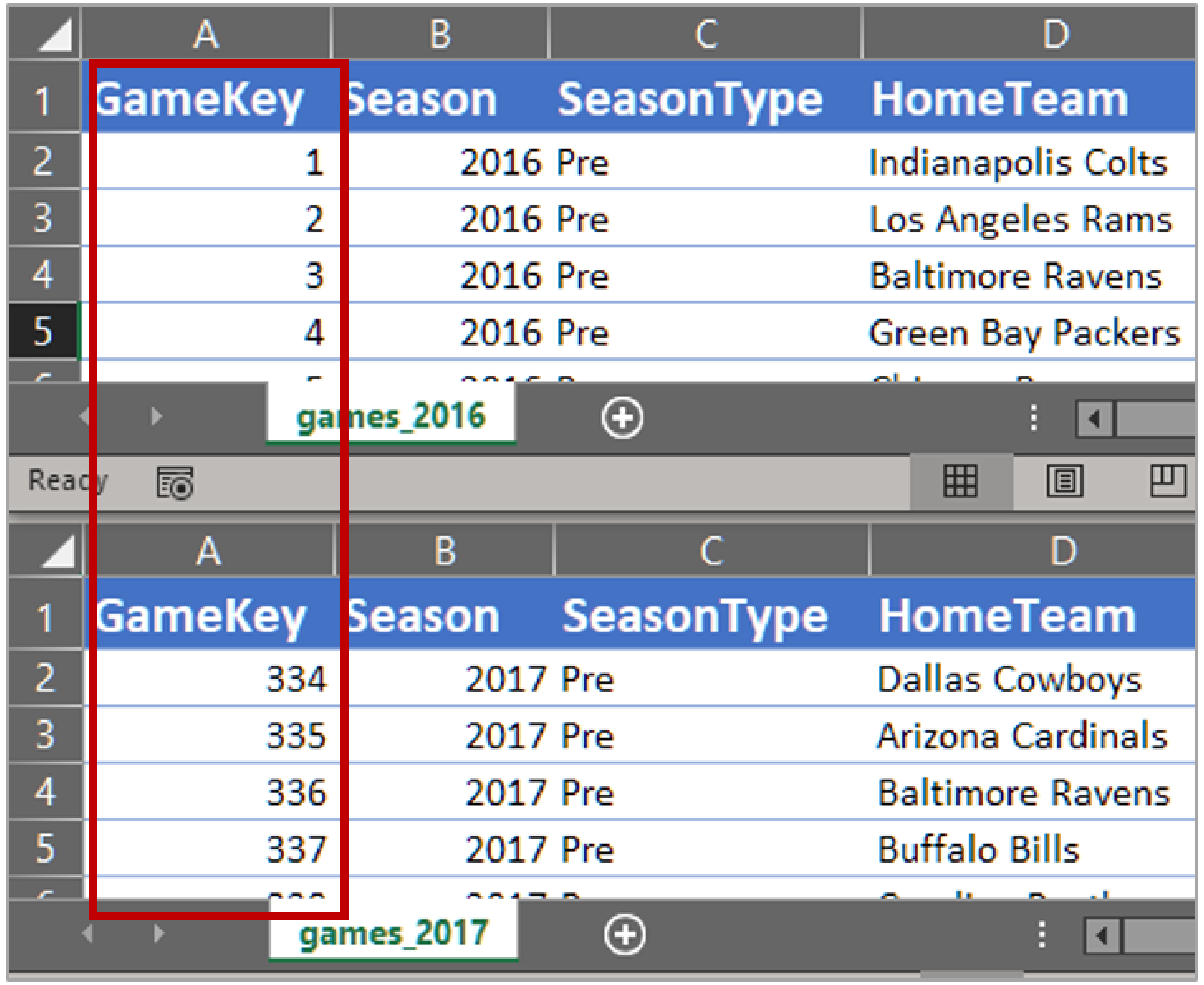


Concatenating rows

- Useful when working with split data

```
pd.concat([df1, df2, ...])
```

- Uses unique key(s) as data frame index
- Includes all rows by default



GameKey	Season	SeasonType	HomeTeam
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GameKey	Season	SeasonType	HomeTeam
334	2017	Pre	Dallas Cowboys
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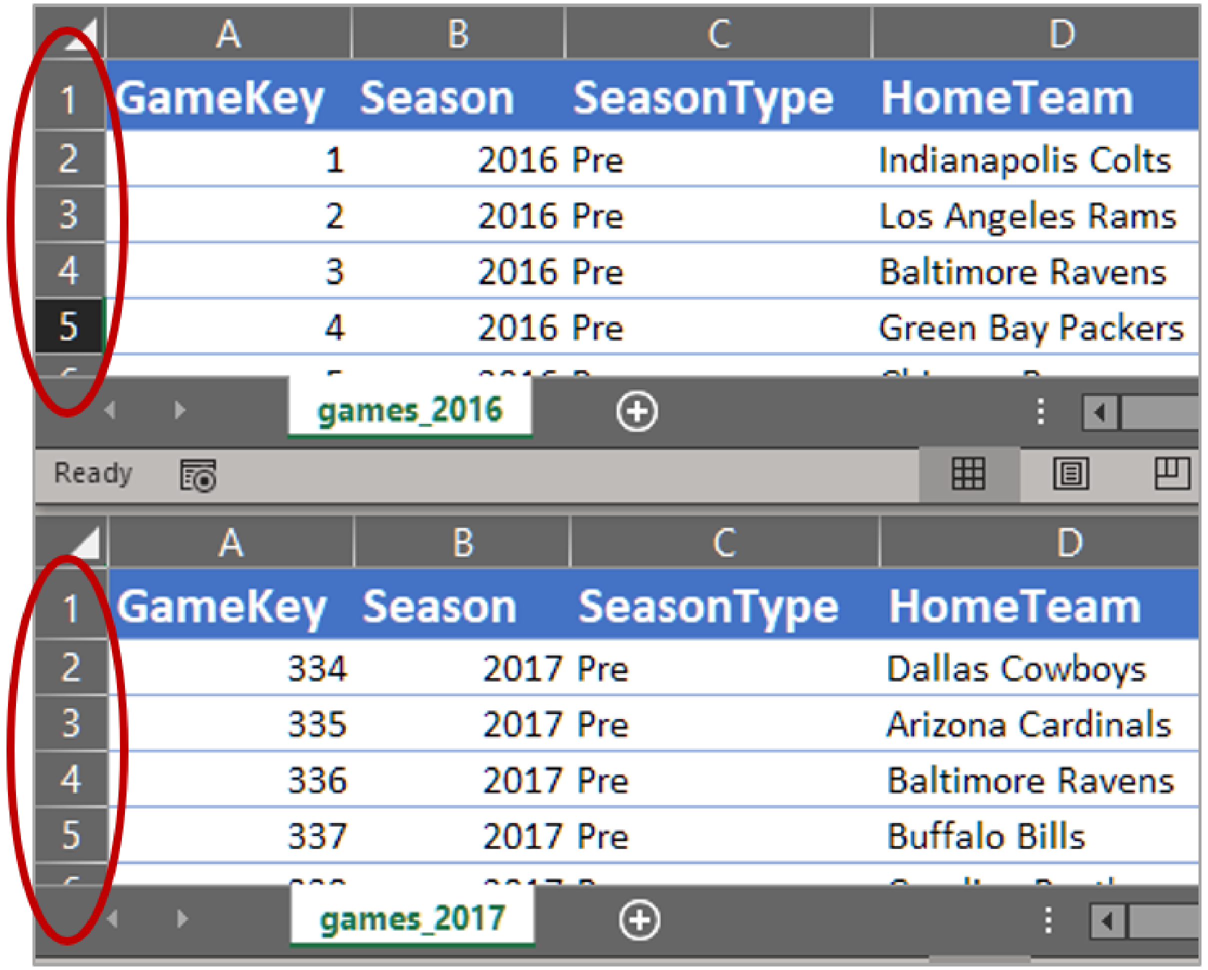
336	2017	Pre	Baltimore Ravens
337	2017	Pre	Buffalo Bills

Concatenating rows with overlapping indices

- Data frame indices may overlap
- Don't worry!

```
pd.concat([df1, df2, ...],
    ignore_index=True)
```

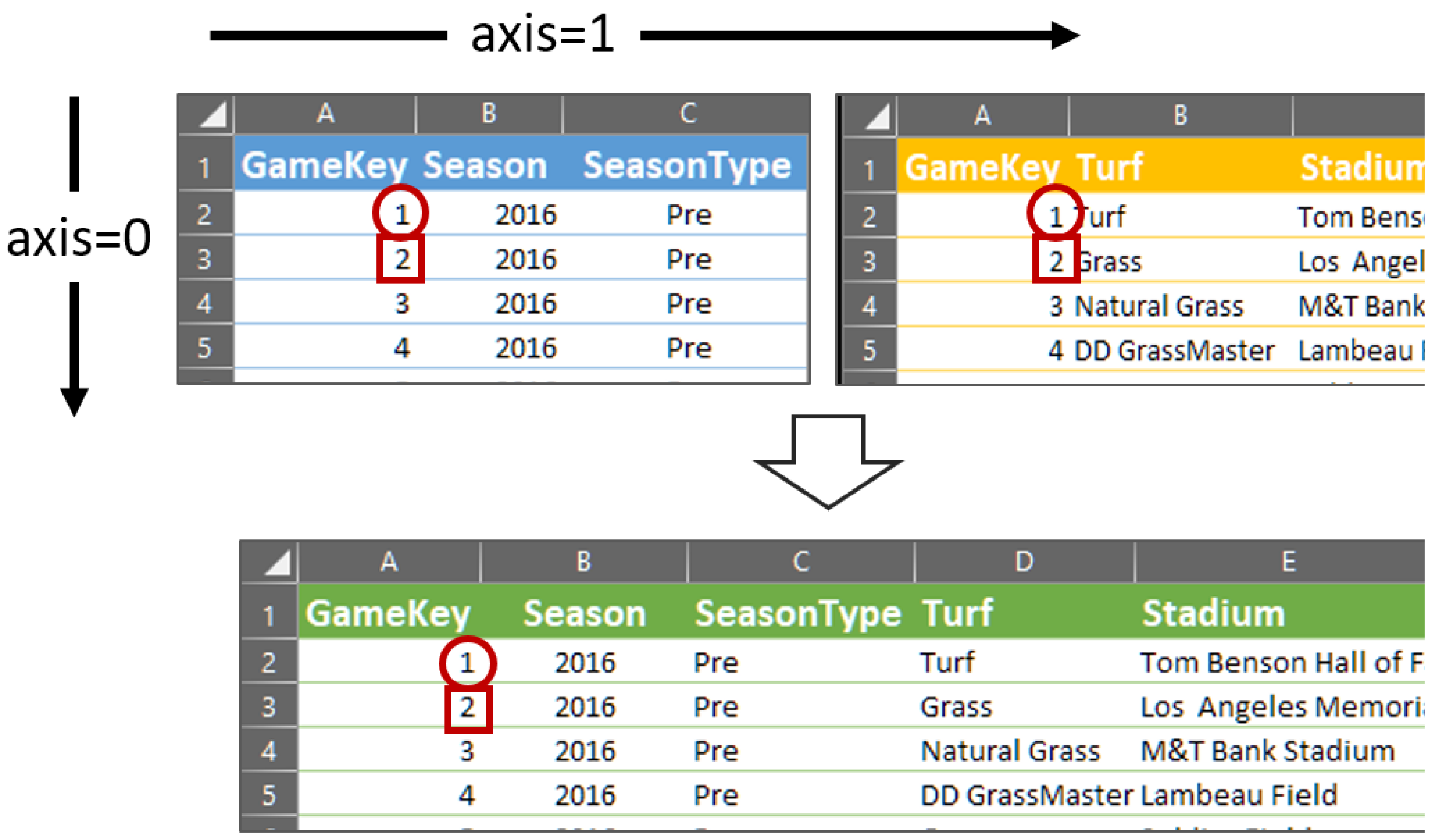


Concatenating columns

- Like pasting tables side by side
- Across columns: axis=1

```
pd.concat([df1, df2, ...],
    axis=1)
```

- Includes all columns by default



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Power and flexibility

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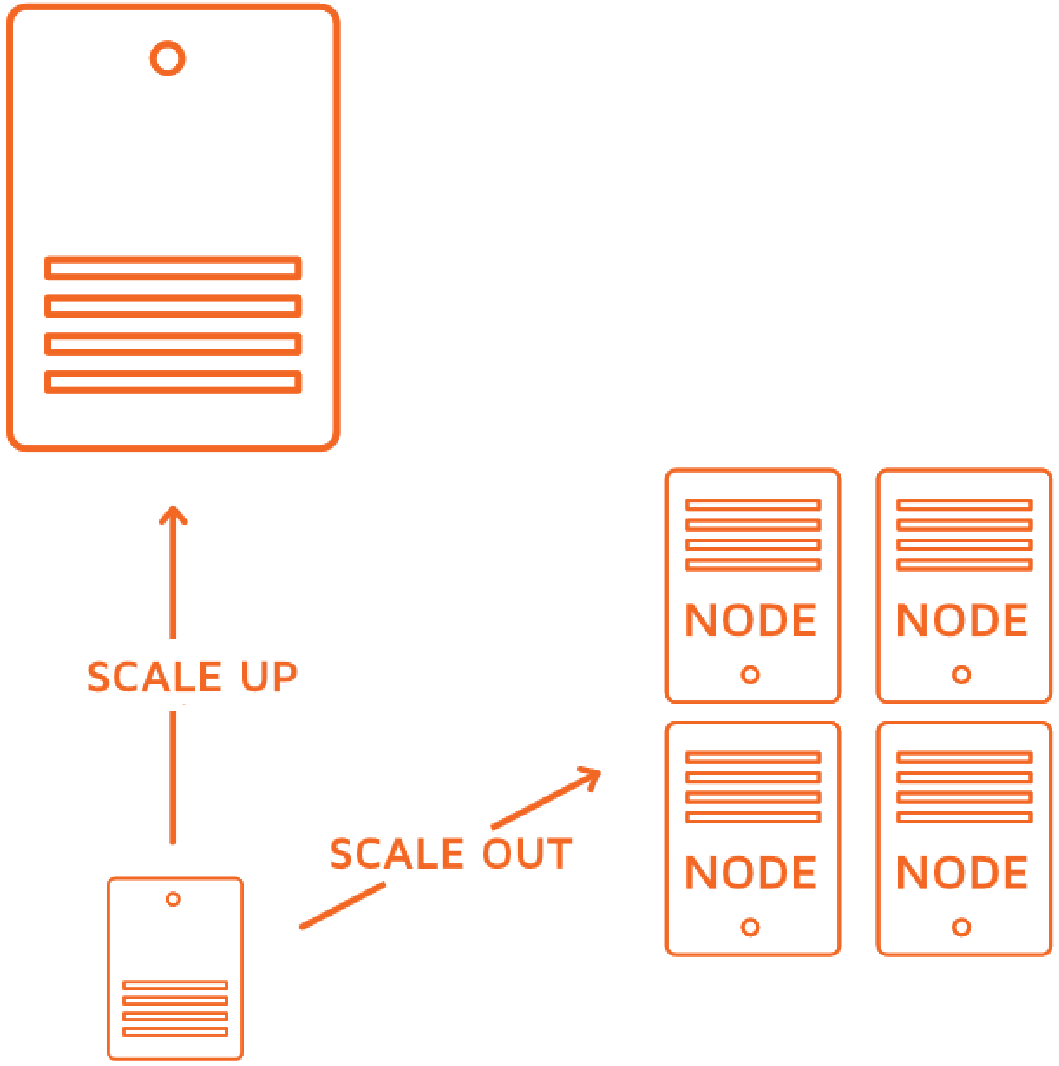
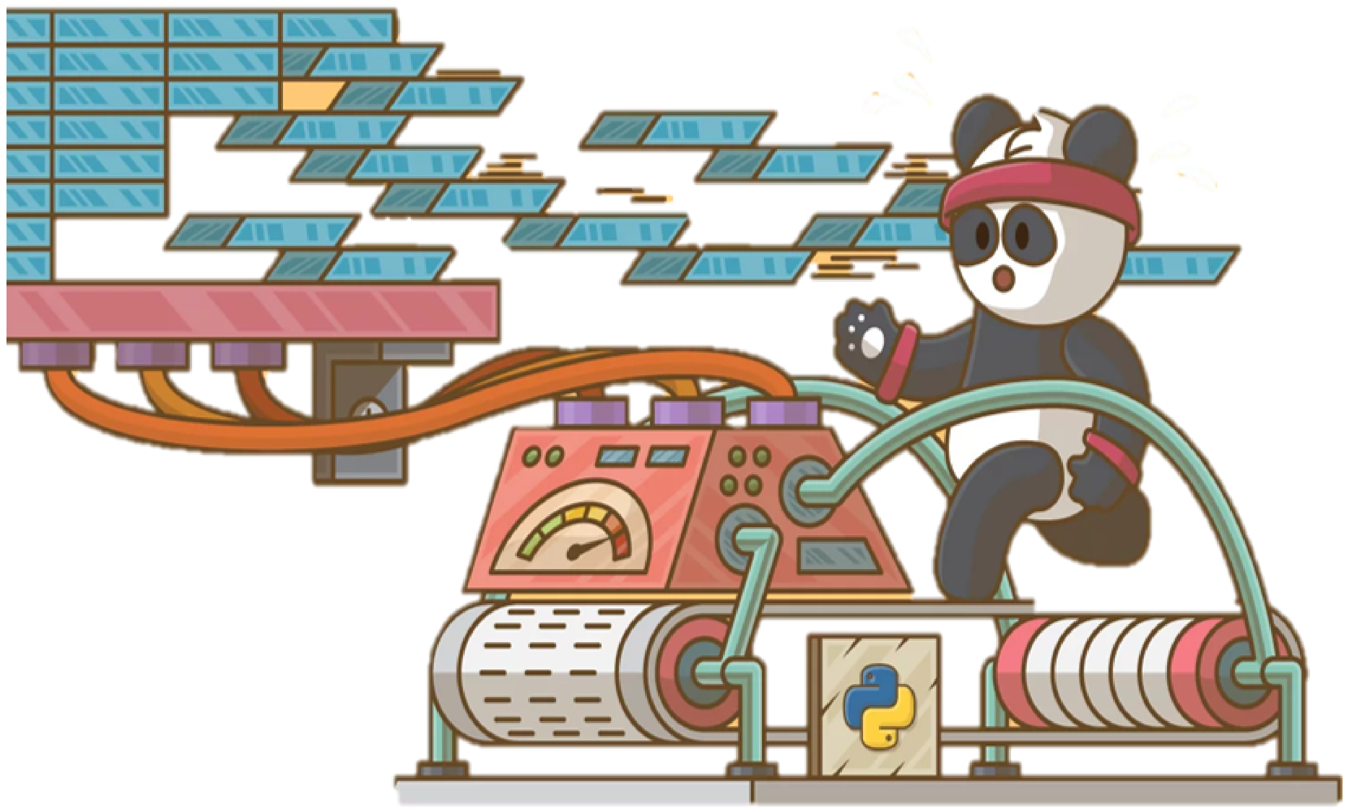
Scalability

- No hard limits on data frame size
- Built-in ways to "chunk" data
- Use distributed/parallel computing



Efficiency

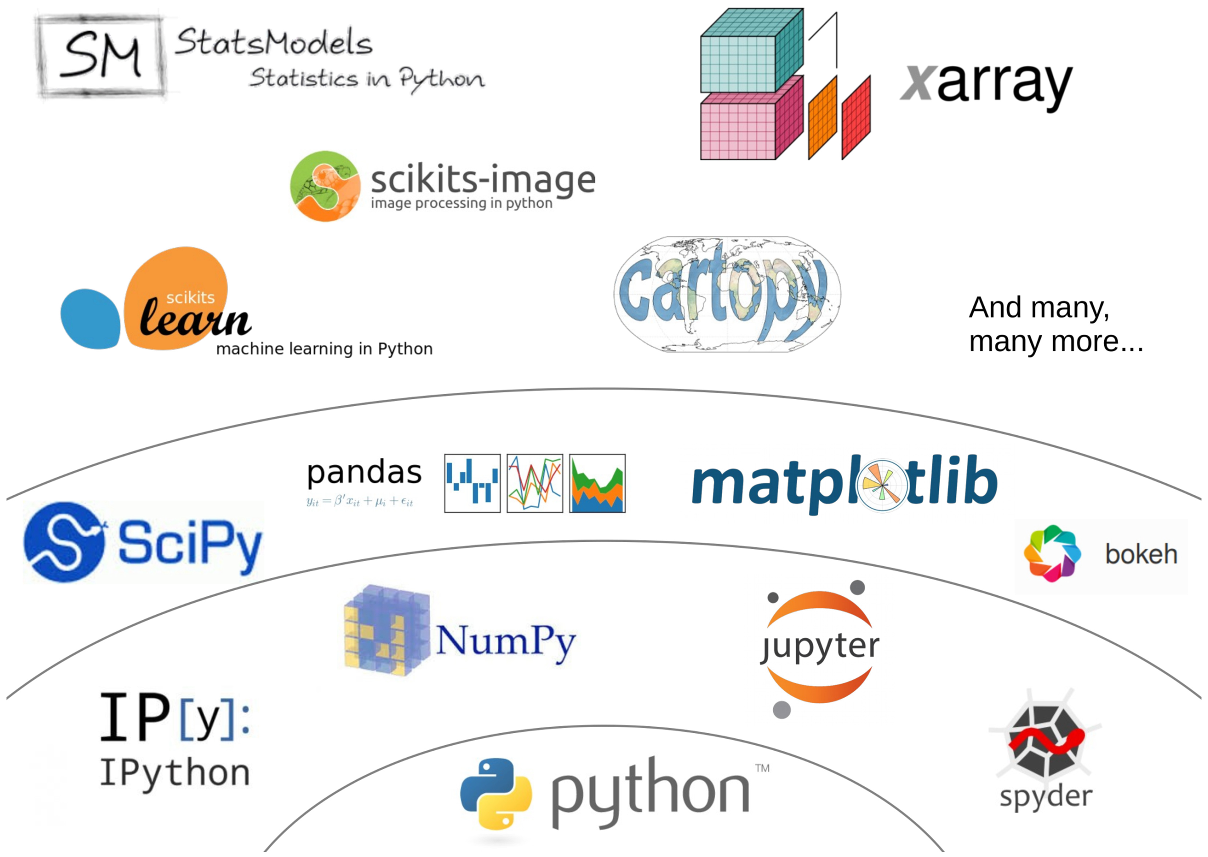


- Join on multiple columns
- Preference for simple code

```
joined_df = left_df.merge(right_df)
```

Integration

- Improved speed and scale
- Data visualization
- Machine learning



A word on advanced spreadsheet usage

- Data models and query tools
- Programming languages
- Advanced formulas

Let's practice!

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